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Informatica® Cloud Application Integration
May 2025

Generate Image with NVIDIA NIM

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Table of Contents

Preface 5

Chapter 1: Introduction to Generate Image with NVIDIA NIM recipe 6

Generate Image with NVIDIA NIM recipe contents 6

Generate Image with NVIDIA NIM recipe assets 7

Chapter 2: Using the Generate Image with NVIDIA NIM recipe 8

Step 1. Copy and access the recipe 8

Step 2. Configure and publish the NVIDIA NIM connection 9

Step 3. Configure and publish the Slack connection 9

Step 4. Configure and publish the processes 10

Step 5. Create webhook in Slack 11

Preface

Use *Generate Image with NVIDIA NIM* to learn how to create images that align with the text-based descriptions provided in natural language prompts. This guide assumes that you have an understanding of the NVIDIA NIM Connector and Slack Connector concepts.

CHAPTER 1

Introduction to Generate Image with NVIDIA NIM recipe

The Generate Image with NVIDIA NIM recipe is based on REST and SOAP APIs. The recipe create images that aligns with the text-based descriptions provided in the natural language prompts and posts them to the Slack channel.

The process is triggered by a message in the Slack channel where the bot is deployed. It sends a prompt to NVIDIA NIM to generate an image, then receives the response in base64 format. The process converts this into an image attachment, uploads the file to Slack, and displays the generated image in the same channel.

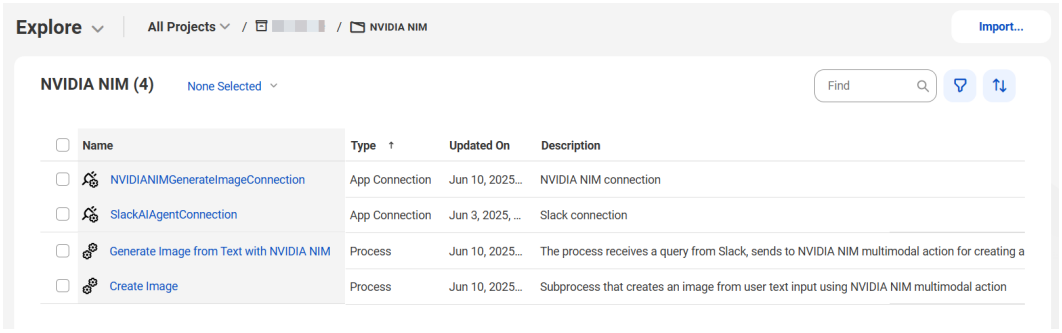
The recipe supports image generation in the following formats, with a maximum size of 5 MB:

`.png, .jpeg, .jpg.`

Generate Image with NVIDIA NIM recipe contents

The Generate Image with NVIDIA NIM recipe contains app connections and processes.

The following image shows the assets that the Generate Image with NVIDIA NIM recipe package contains:



Generate Image with NVIDIA NIM recipe assets

The following table lists the assets that the Generate Image with NVIDIA NIM recipe package contains:

Asset Name	Asset Type	Description
NVIDIANIMGenerateImageConnection	App connection	NVIDIA NIM connection.
SlackAIAgentConnection	App connection	Slack connection.
Create Image	Process	Subprocess that creates an image from user text input using NVIDIA NIM multimodal action.
Generate Image from Text with NVIDIA NIM	Process	The process receives a query from Slack, sends to NVIDIA NIM multimodal action for creating an image from a description, and returns the generated image to the Slack channel.

CHAPTER 2

Using the Generate Image with NVIDIA NIM recipe

To use the Generate Image with NVIDIA NIM recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the NVIDIA NIM connection.
3. Configure and publish the Slack connection.
4. Configure and publish the processes.
5. Create webhook in Slack.

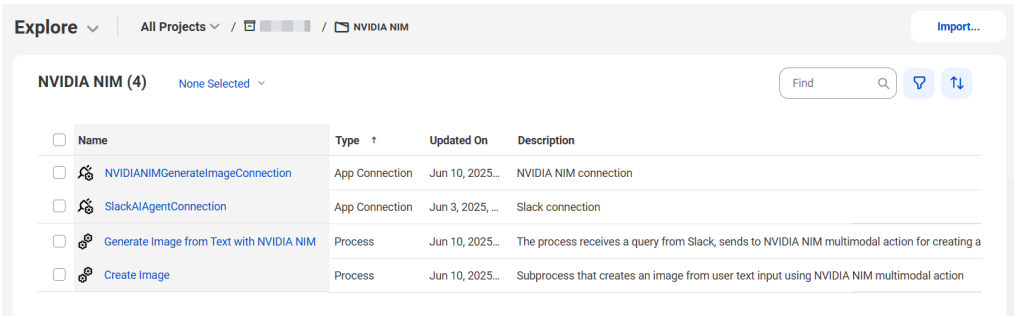
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Generate Image with NVIDIA NIM** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the NVIDIA NIM connection

Configure the API key to authenticate NVIDIA NIM API requests, and then publish the NVIDIANIMGenerateImageConnection connection.

1. Open the **NVIDIANIMGenerateImageConnection** connection.
2. From the **Type** list, select **NVIDIA NIM**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter the API key in the **Api_Key** property. The **Api_Key** property authenticates NVIDIA NIM API requests.
5. Save and publish the connection.

Step 3. Configure and publish the Slack connection

Configure the base URL to access the Slack instance, and then publish the SlackAIAgentConnection connection.

1. Open the **SlackAIAgentConnection** connection.
2. From the **Type** list, select **Slack**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. From the **Authentication Type** list, select **OAuth2** and enter values for the following properties:

Property	Description
Grant Type	Required. Select the required grant type based on your use case.
Authorization URL	Required. Enter the OAuth authorization URL for the OAuth provider to authorize the user request. For example: `https://slack.com/oauth/v2/authorize`
Token Request URL	Required. Enter the OAuth token request URL that handles token requests. For example: `https://slack.com/api/oauth.v2.access` The expiration of the refresh token depends on the OAuth provider. You must re-authenticate and publish the connection before the token expires. **Note:** The refresh token doesn't apply to agents and agent groups.
Client ID	Required. Specify the identifier value from the OAuth provider.
Client Secret	Required. Enter the client secret to connect to the OAuth provider.
Scope	Required. Specify the scope. The scope in OAuth authentication limits an application's access to a user's account. You can select multiple scopes for a single client. To enter multiple scopes, separate each value with a space.

Property	Description
Session Duration	Optional. Enter the number of minutes to wait before refreshing the session. This is used if you cannot obtain the token expiration from the OAuth provider. Default is 60 minutes.
Other Parameters	Optional. Enter parameters to customize the authorization request, separated with semicolons. You can provide extra information or specify requirements of the OAuth provider.
Authorization Status	Optional. Indicates the current status, the name of the user who authorized, and the last time when the authorization was completed.
Authorize Access	Click to initiate the authorization workflow using OAuth.

5. In the **Connection Properties** section, enter the base URL in the **Base_URL** property. The base URL connects to the Slack API and sets the primary endpoint for all Slack requests.
6. Save and publish the connection.

Step 4. Configure and publish the processes

Configure the image settings and authorization access, and then publish the processes.

1. Open the **Create Image** process.
2. Optionally, on the **Assignments** tab of the **Prepare Request PO** step, update the image settings in the **Request** field using the Expression Editor, as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<height m:type="xs:int">1024</height>
<width m:type="xs:int">1024</width>
<text_prompts m:isArray="true">
<text>{$input.User_Input}</text>
<weight m:type="xs:double">1</weight>
</text_prompts>
<cfg_scale m:type="xs:double">5</cfg_scale>
<clip_guidance_preset>NONE</clip_guidance_preset>
<sampler>K_EULER_ANCESTRAL</sampler>
<samples m:type="xs:int">1</samples>
<seed m:type="xs:int">0</seed>
<steps m:type="xs:int">25</steps>
<style_preset>none</style_preset>
</root>
```

3. Optionally, on the **Assignments** tab of the **Set Answer** step, update the image name and format in the **response** field. By default, the process receives a response from NVIDIA NIM action in base64 format and creates an attachment named `picture.jpeg` with the format `image/jpeg`. You can use this default format or change it to any format supported by Application Integration.
4. Save and publish the process.
5. Open the **Generate Image from Text with NVIDIA NIM** process.
6. On the **Start** tab of the **Start** step, enter the user in the **Allowed Users** field or select **Allow anonymous access**.
7. Save and publish the process.

8. On the **Actions** menu, click **Properties Detail** and copy the service URL.

   You will need the service URL to configure the webhook in Slack. For more information, see "Step 5. Create webhook in Slack" on page 11.

Step 5. Create webhook in Slack

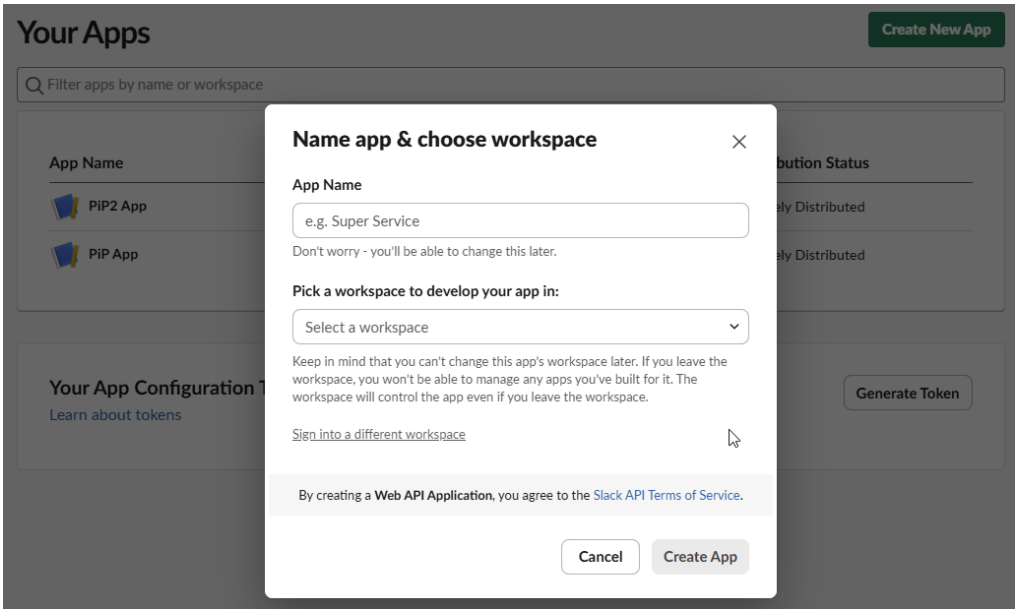
Configure a Slack app that calls an API when a new message arrives.

To create a webhook in Slack, perform the following steps:

1. "Create a Slack App" on page 11
2. "Enable the events API" on page 11
3. "Subscribe to the event" on page 12
4. "Set the required OAuth scopes" on page 13
5. "Enable token rotation" on page 14
6. "Add bot to a channel" on page 15

Create a Slack App

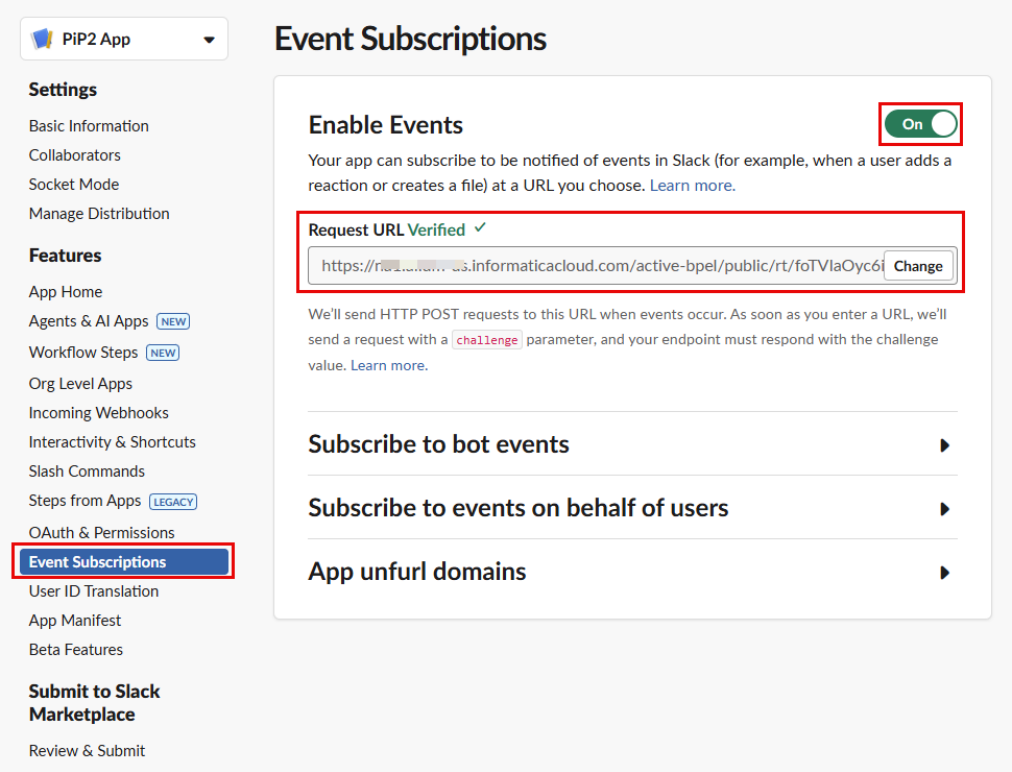
1. Go to `https://api.slack.com/apps` and log in to your account.
2. Click **Create New App** > **From scratch**.
3. Enter the app name and select your workspace, and then click **Create App** as shown in the following image:



Enable the events API

1. From the navigation bar, go to **Event Subscriptions**.
2. Turn on the toggle switch to enable events.

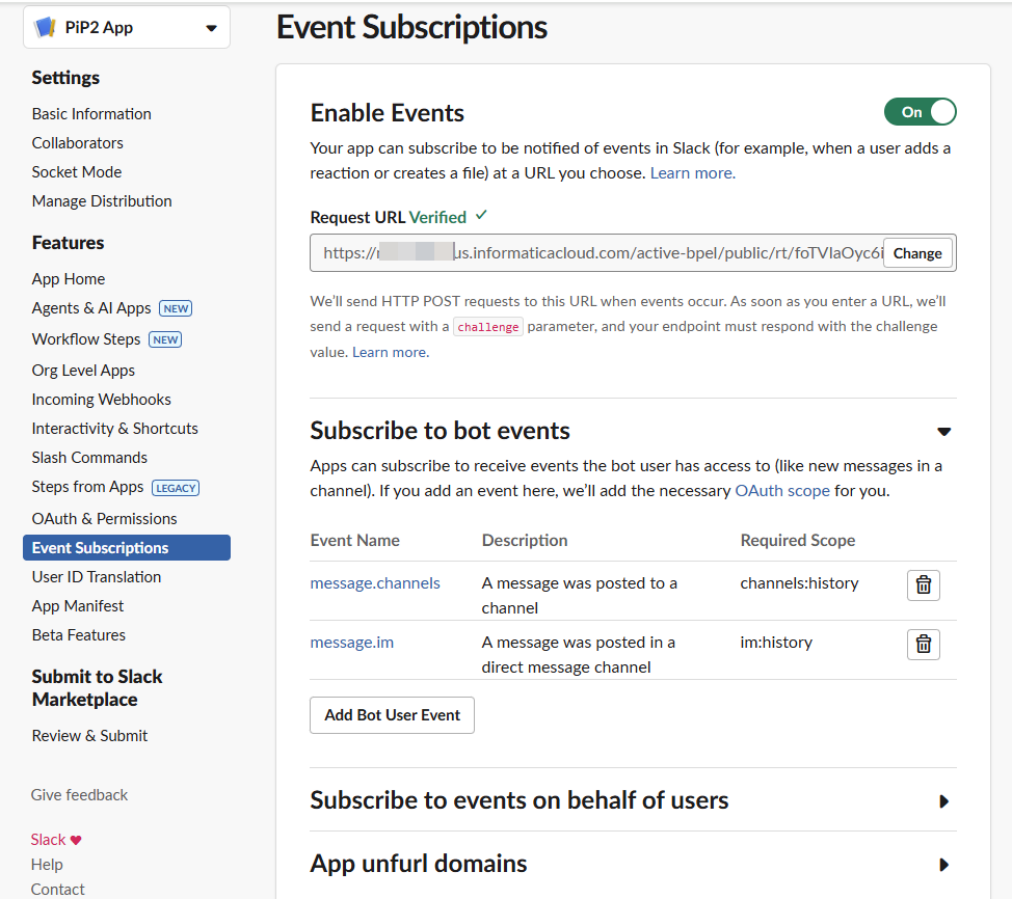
3. In the **Request URL** field, enter the service URL that you get after publishing the process. Slack sends a `challenge` request to verify the URL as shown in the following image:



Subscribe to the event

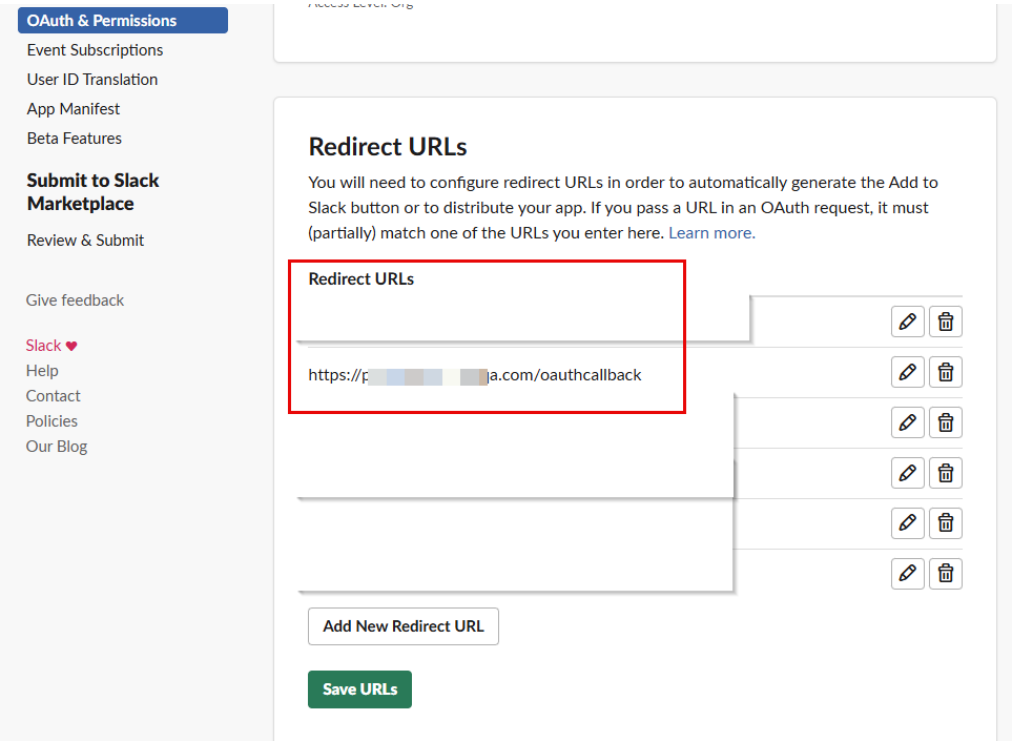
1. In the **Subscribe to bot events** section, click **Add Bot User Event** and add one of the following events as required:
   - `message.channels` - to receive messages from public channels.
   - `message.groups` - for private channels.

   The event allows your app to receive messages from channels it belongs to as shown in the following image:



Set the required OAuth scopes

1. Go to **OAuth & Permissions**.
2. In the **Bot Token Scopes** section, click **Add an OAuth Scope** and add one of the following OAuth scopes:
   - `channels:history` — to read public channel messages.
   - `groups:history` — to read messages from private channels, if needed.
   - `chat:write` — if your bot needs to send messages.
3. In the **Redirect URLs** section, click **Add New Redirect URL** and add the redirect URLs in `<Cloud Application Integration POD URL>/oauthcallback` format and save the URLs. For example, `https://na1.dm-us.informaticacloud.com/oauthcallback` as shown in the following image:

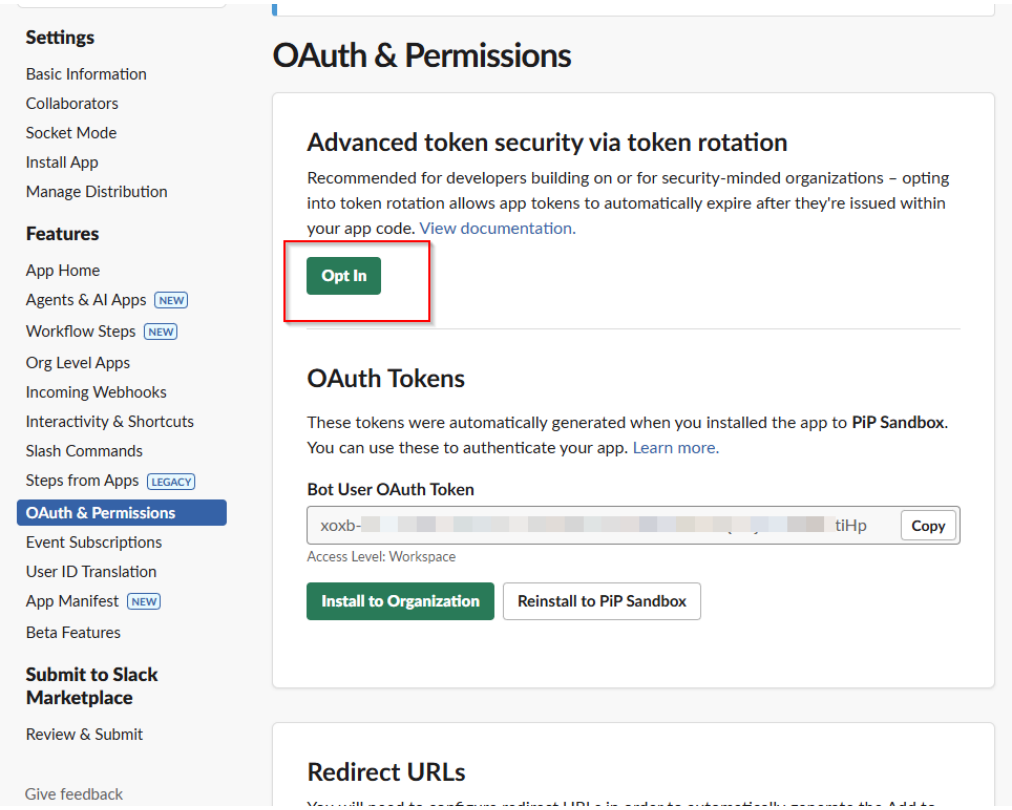


4. Install the app using the **Install App** section.

Enable token rotation

1. Go to **OAuth & Permissions**.

2. In the **Advanced token security via token rotation** section, click **Opt In**.



Add bot to a channel

1. In Slack, go to the required channel.
2. Invite the bot using `/invite @<your-bot-name>`.